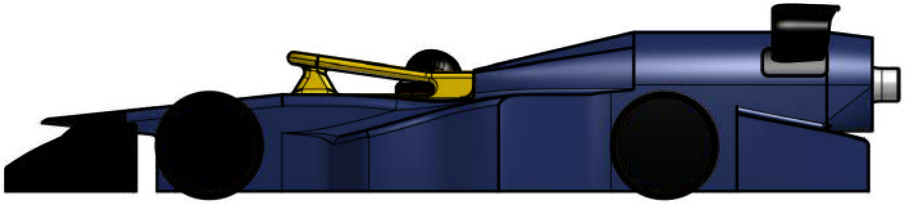
ENGINEERING DRAWINGS AERIS
2025 WORLD FINALS

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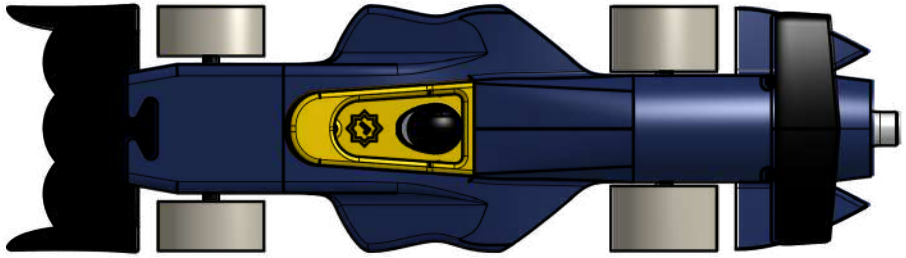
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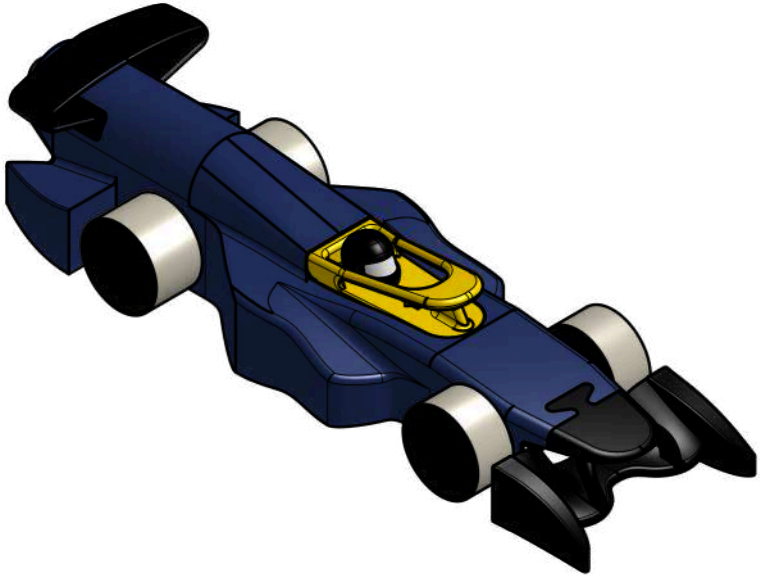
onshape · Zeno Watch Basel · Crédit Mutuel · Rotary · Lycée Jean-Mermoz · Saint-Louis Agglomération Terres d'avenir · Lions International · Valoria Communiquer par l'objet · Alsace Dépannage · Sofitex De l'intérim au CDI · Productec Logiciels et services de programmation CNC · Biederthal · Ansys · Guerra Frères Construction · ISL · helbling · SunChemical a member of the DIC group · dic Color & Comfort · Kanzen Roulement.com · ClickUp · Leihouse.com



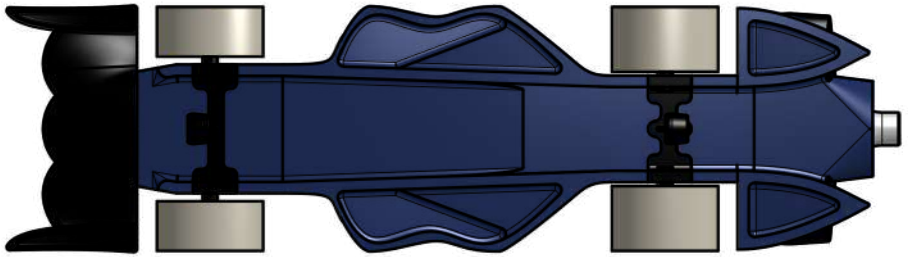
Right



Top



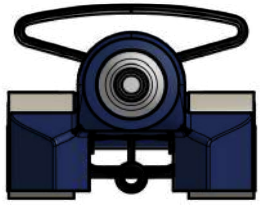
Isometric



Bottom



Front



Back

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

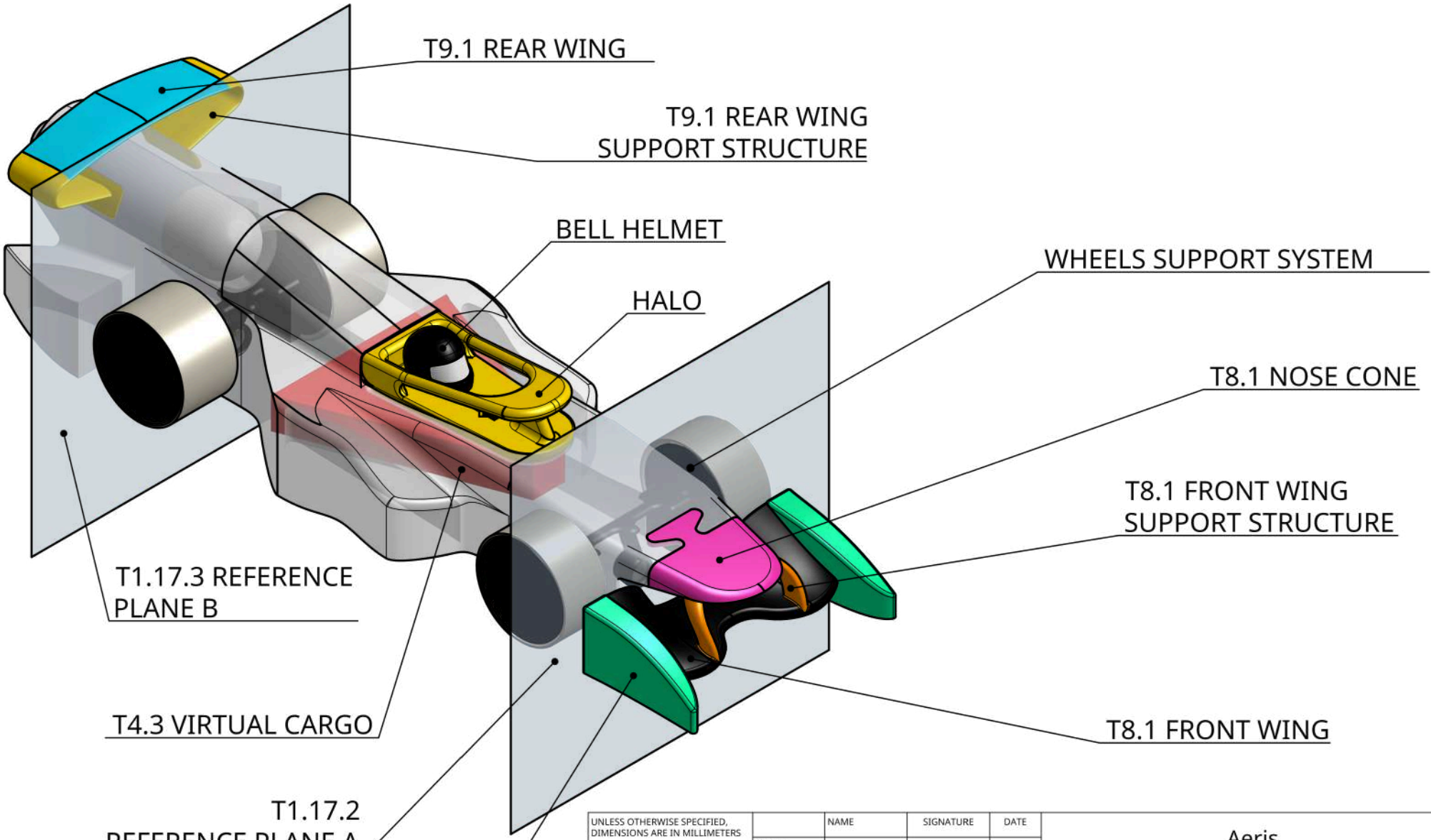
Aeris

TITLE

Fully Assembled Car

SIZE	DWG NO.	REV.
A4		

SCALE	WEIGHT	SHEET
1:2		1 of 12



		NAME	SIGNATURE	DATE
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS				
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

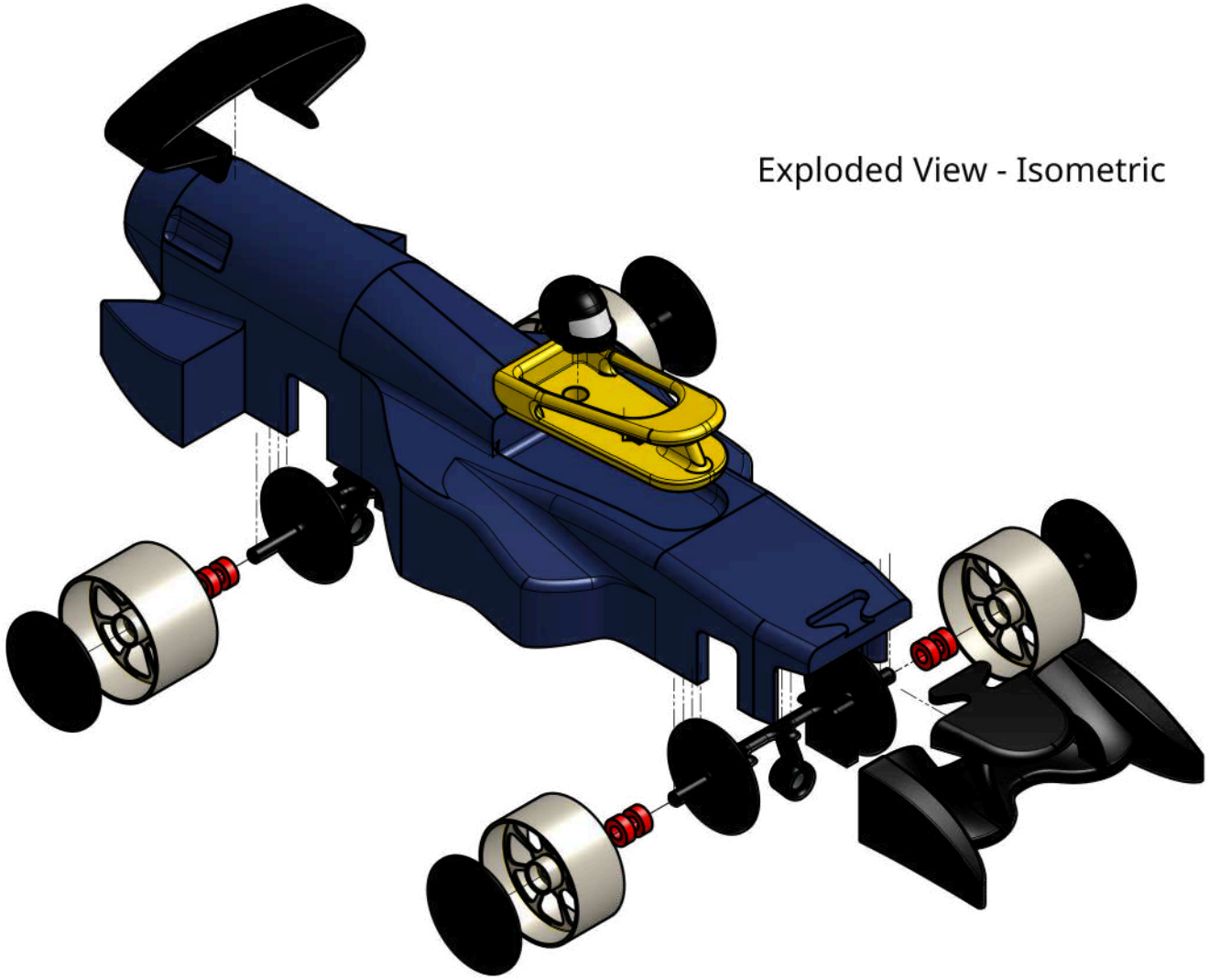
Aeris

TITLE: Component Identification

SIZE	DWG NO.	REV.
A4		

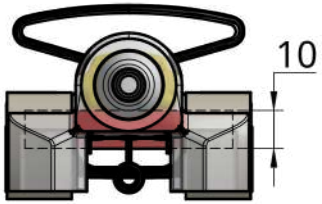
SCALE	WEIGHT	SHEET
1:1.2		2 of 12

Name	Material	Mass	Qty
Halo	PLA 5%	3,965 g	1
Bell Helmet	PLA 5%	1,043 g	1
Chassis	Official Denford Block	20,07 g	1
Tether Line Guides	Ceramic	0.067 g	2
Hybrid Bearings	Steel and Ceramic	0.217 g	8
Front Wheel	Aluminum 7075	2,591 g	2
Rear Wheel	Nylon PA12	3,174 g	2
Front Axle	Nylon PA12	2,369 g	2
Rear Axle	Nylon PA12	2,593 g	
Flask	PLA	0,52 g	4
Front Wing	Nylon PA12	5,223 g	1
Rear Wing	Nylon PA12	3,192 g	1

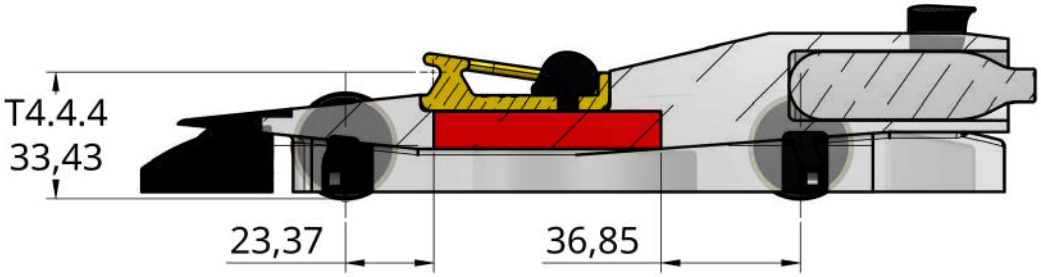


Exploded View - Isometric

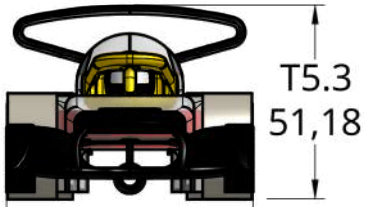
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE	Aeris
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03	
SURFACE FINISH	CHECKED				TITLE: Car Exploded View
	APPROVED				
DO NOT SCALE DRAWING					
BREAK ALL SHARP EDGES AND REMOVE BURRS					SIZE A4 · DWG NO. · REV.
FIRST ANGLE PROJECTION	MATERIAL		FINISH		SCALE 1:1.4 · WEIGHT · SHEET 3 of 12



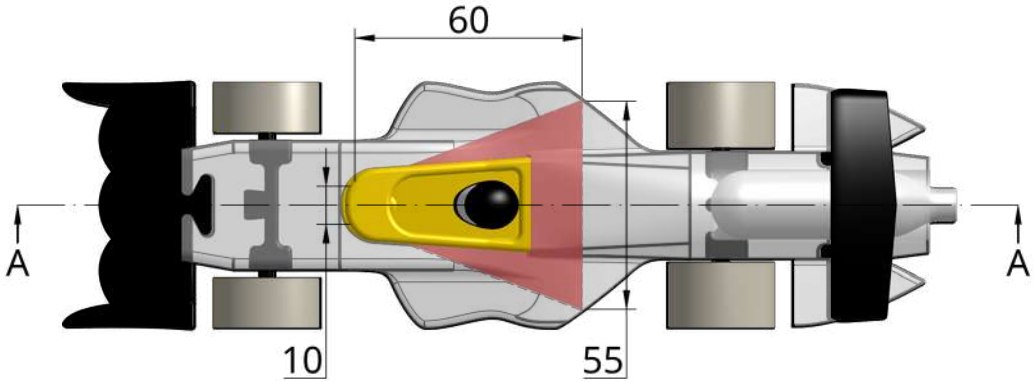
Back



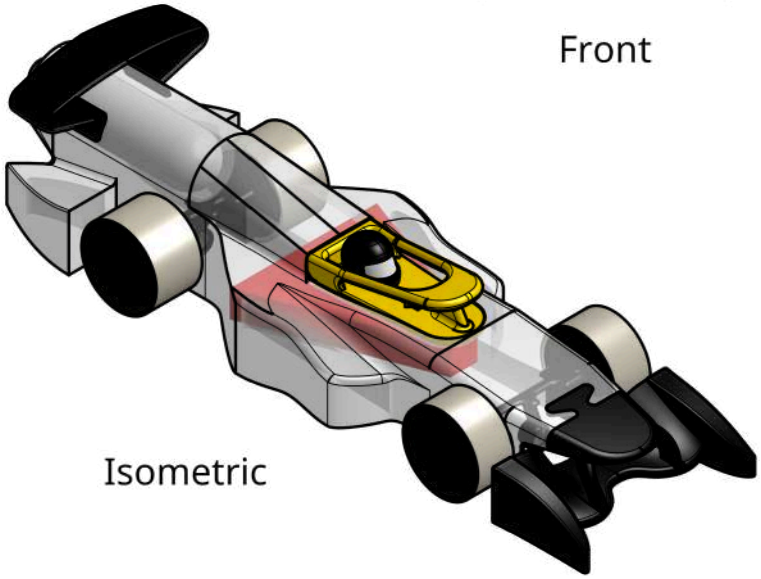
A - A



Front



Top



Isometric

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

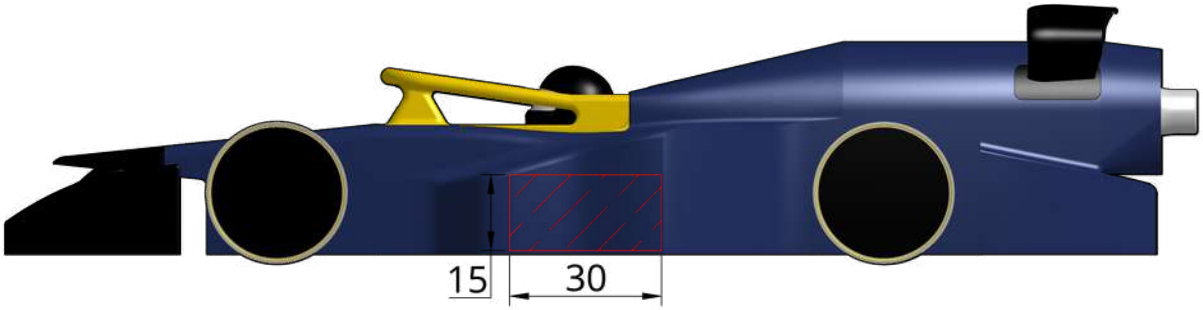
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TITLE

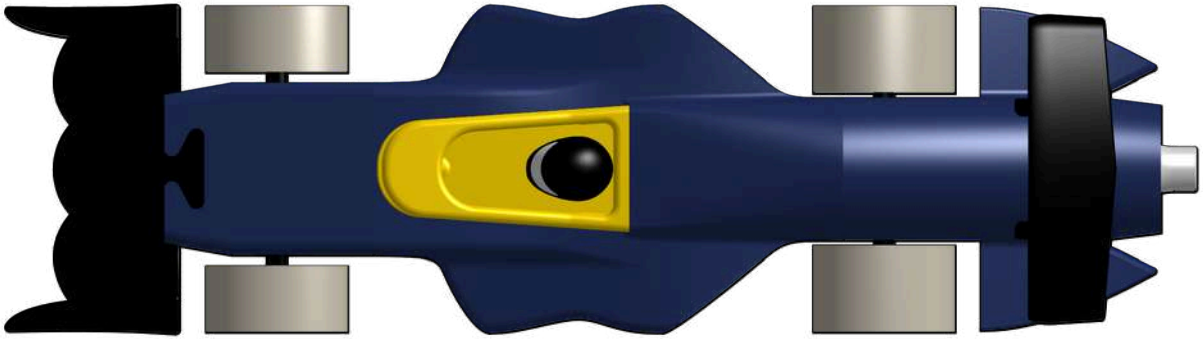
T4.3 Virtual Cargo Identification

SIZE	DWG NO.	REV.
A4		

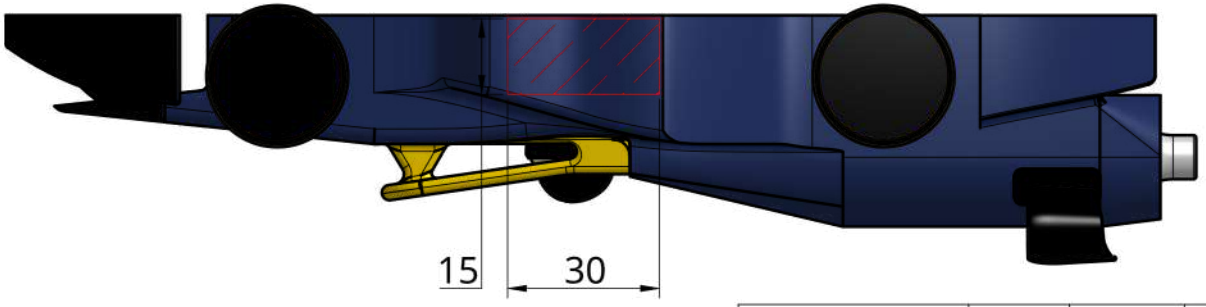
SCALE	WEIGHT	SHEET
1:2		4 of 12



Right



Top



Left

T1.12 STEM Racing™ logo decal

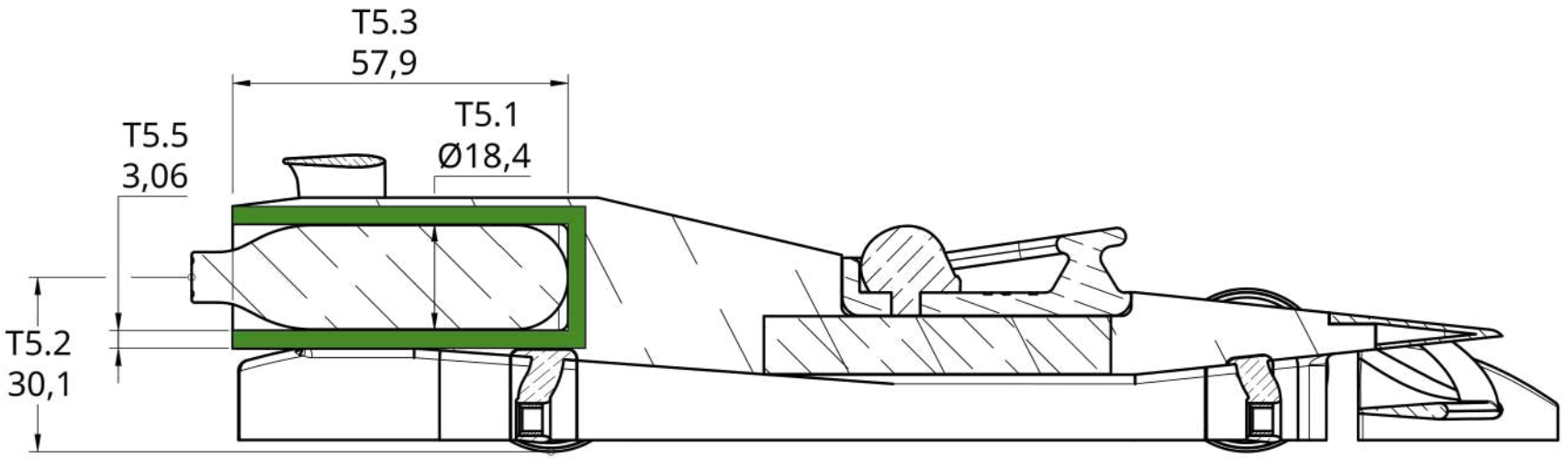
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH √	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

Aeris

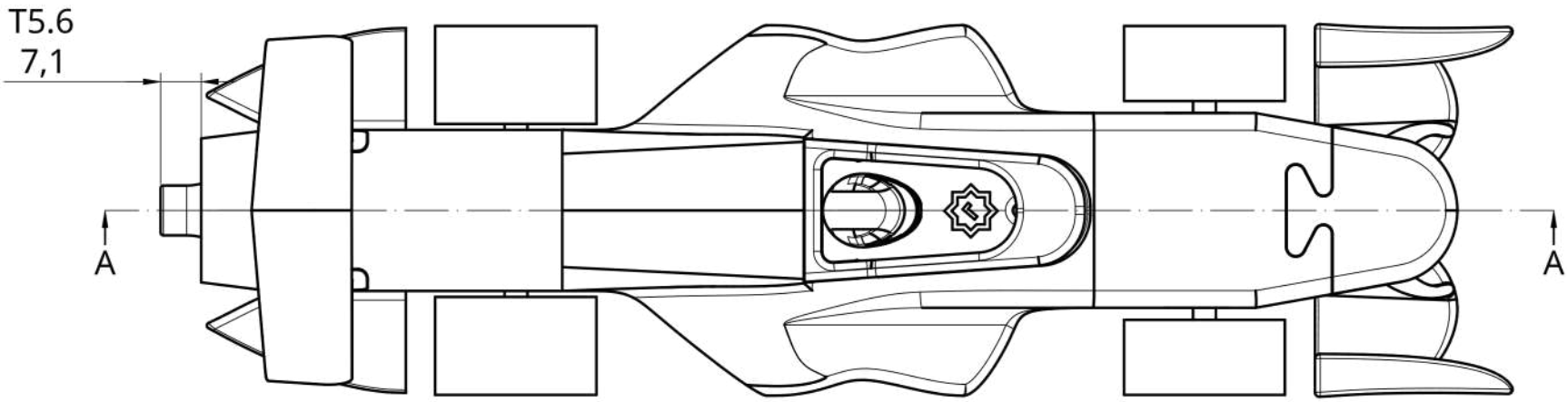
TITLE

Location of official STEM Racing decals

SIZE A4	DWG NO.	REV.
SCALE 1:1.5	WEIGHT	SHEET 5 of 12



A - A



T5.5 Chamber Safety Zone

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

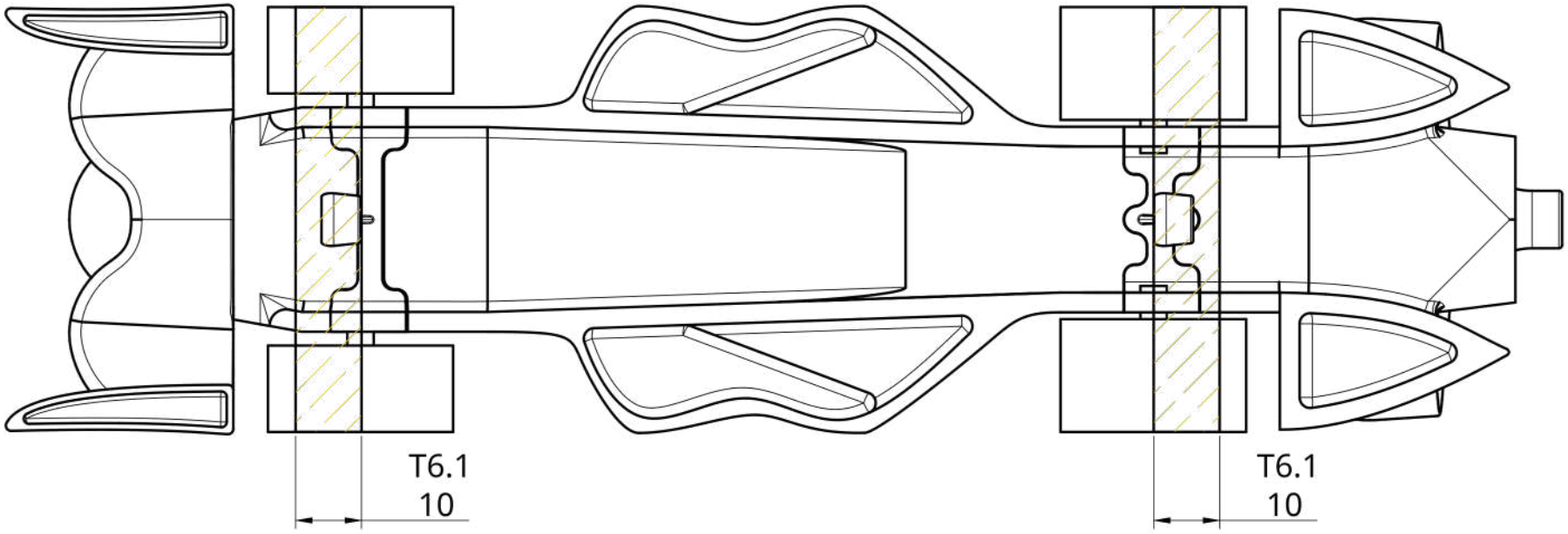
Aeris

TITLE

Power Unit Cartridge

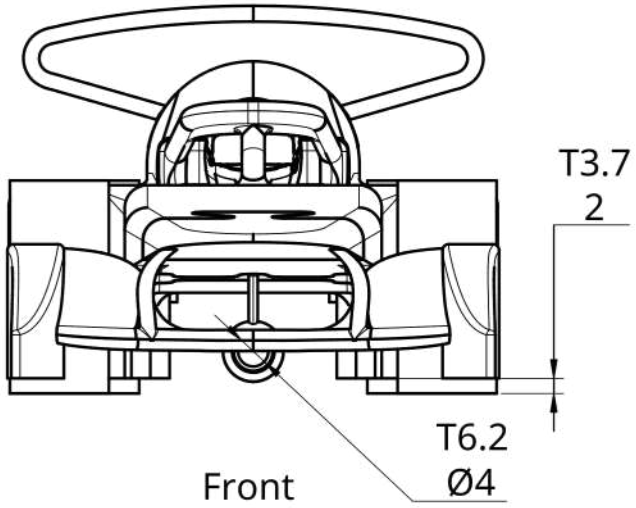
SIZE	DWG NO.	REV.
A4		

SCALE	WEIGHT	SHEET
1:1.2		6 of 12



Bottom

T6.1 Tether Line Guides Location



Front

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

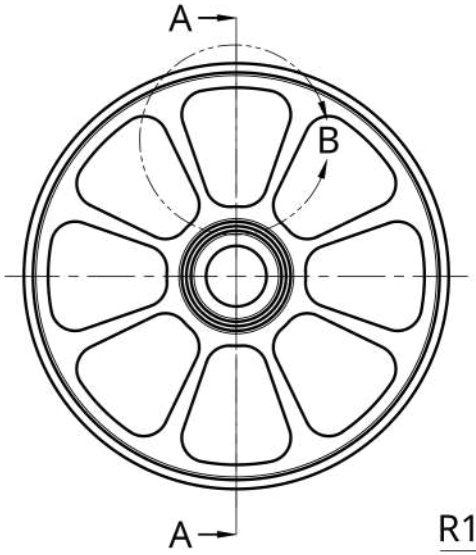
Aeris

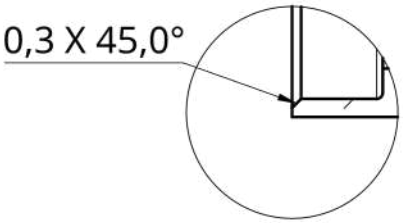
TITLE

Tether Line Guides

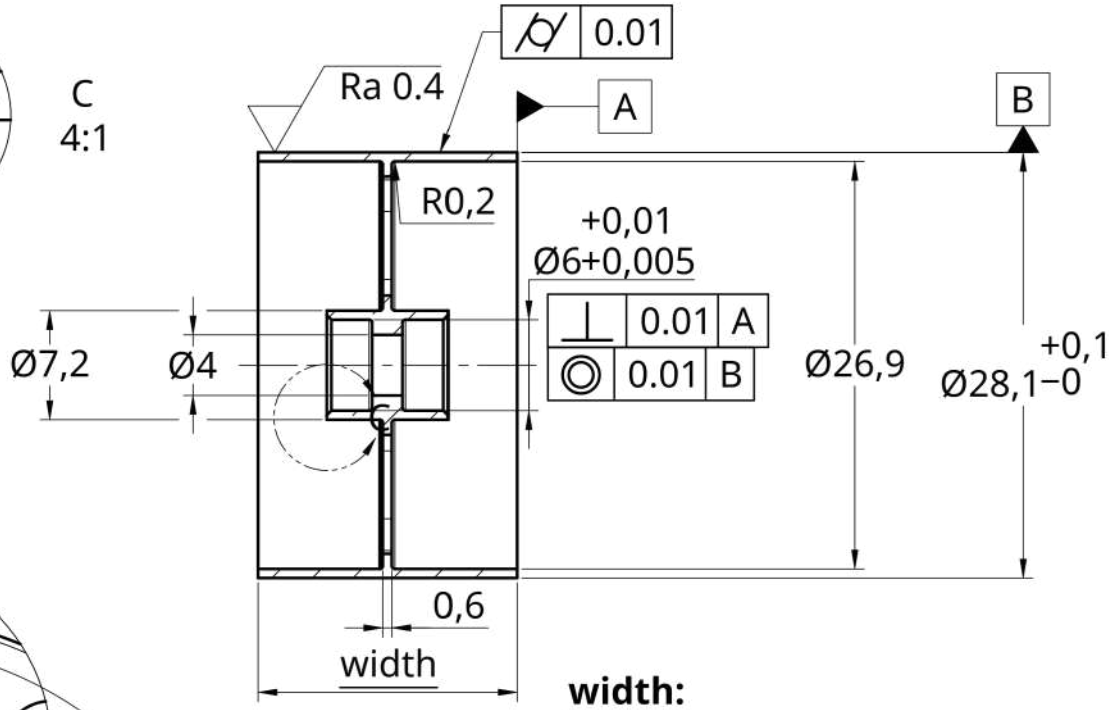
SIZE	DWG NO.	REV.
A4		

SCALE	WEIGHT	SHEET
1:1		7 of 12



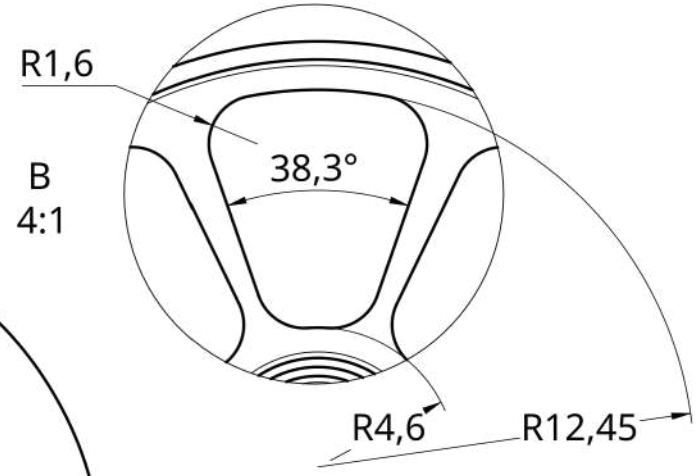


C
4:1

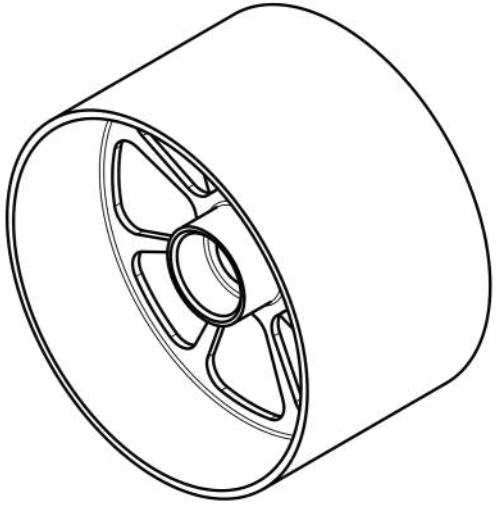


A - A

width:
Front wheel: 13.1 +0/+0.1
Rear wheel: 17.1 +0/0.1



B
4:1



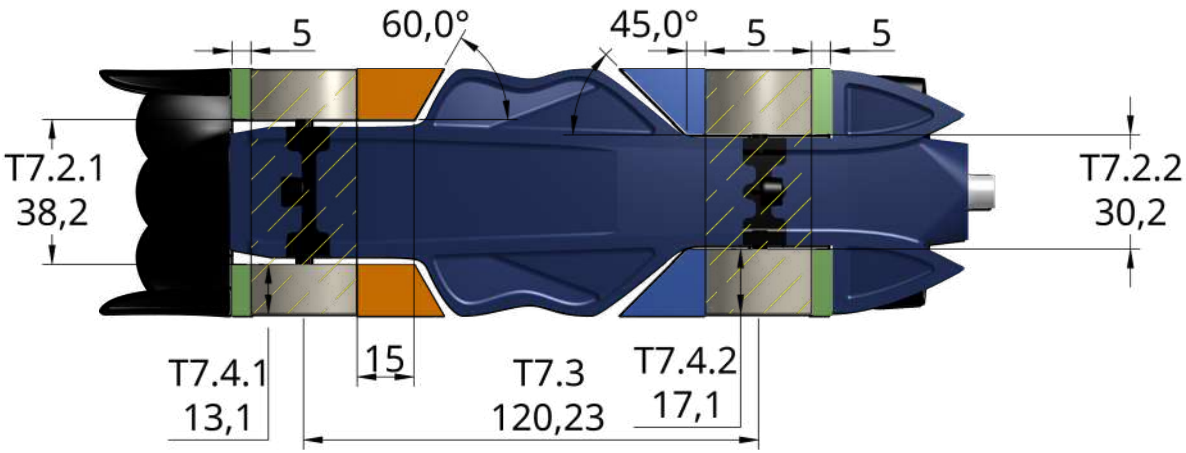
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ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

Aeris

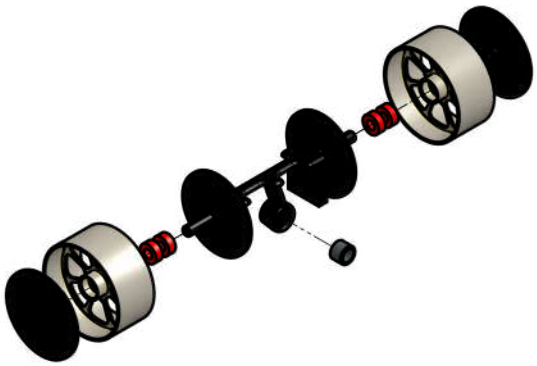
TITLE
Front and Rear Wheels

SIZE	DWG NO.	REV.
A4		

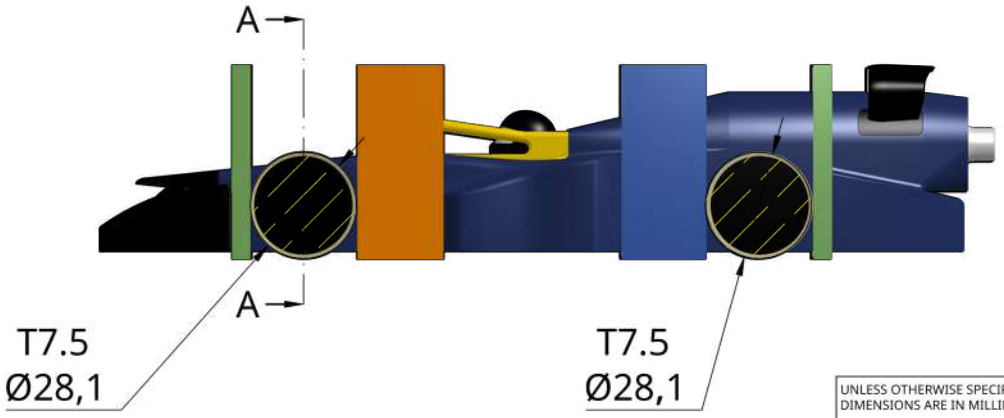
SCALE	WEIGHT	SHEET
2:1		8 of 12



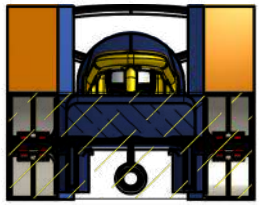
Bottom



Isometric - Exploded System
Front Wheels System



Right



A - A

T7.9 Top and Bottom Visibility

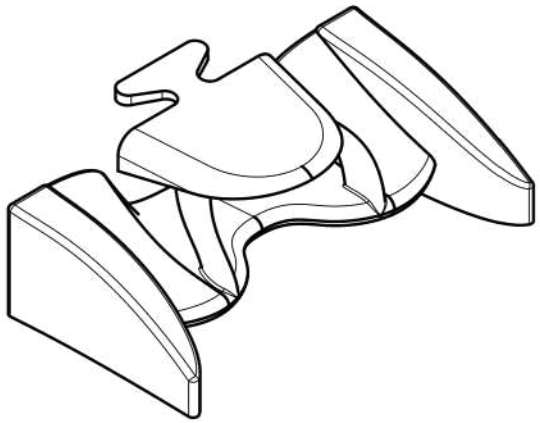
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

Aeris

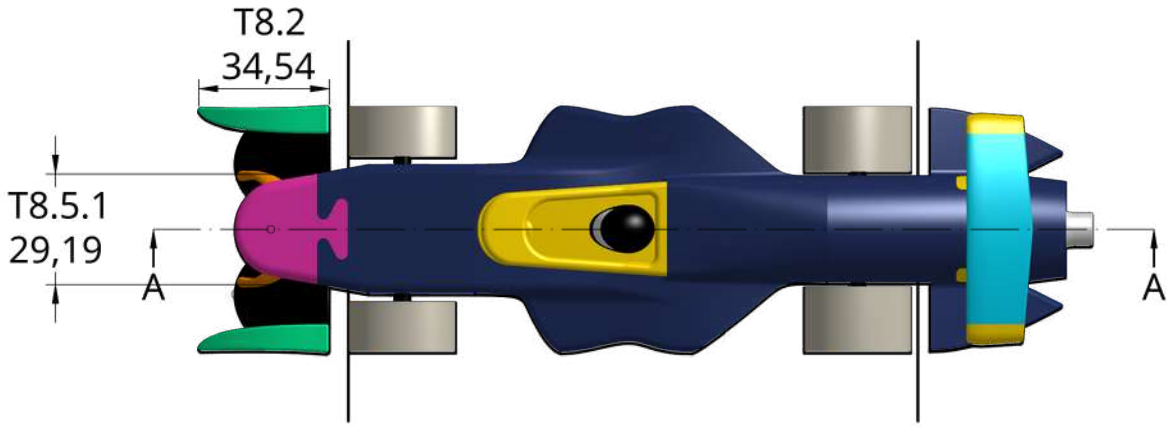
TITLE: Wheels System

SIZE	DWG NO.	REV.
A4		

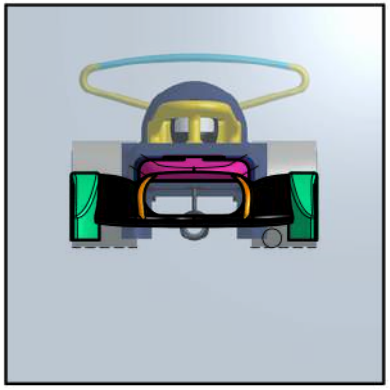
SCALE	WEIGHT	SHEET
1:2		9 of 12



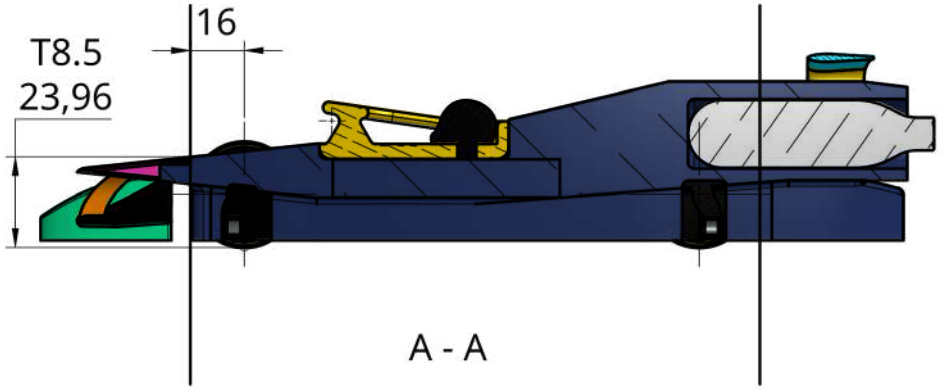
Front Wing - Isometric
1:1



Top



Front



A - A

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

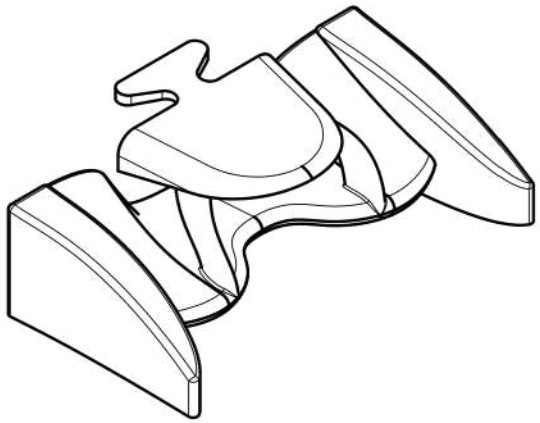
Aeris

TITLE

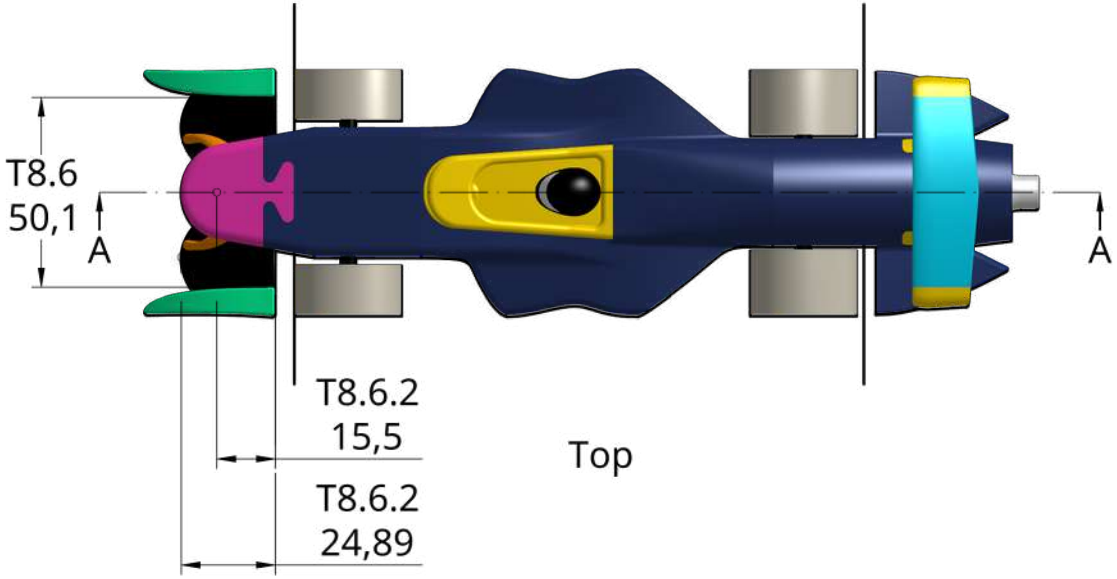
Nose cone

SIZE	DWG NO.	REV.
A4		

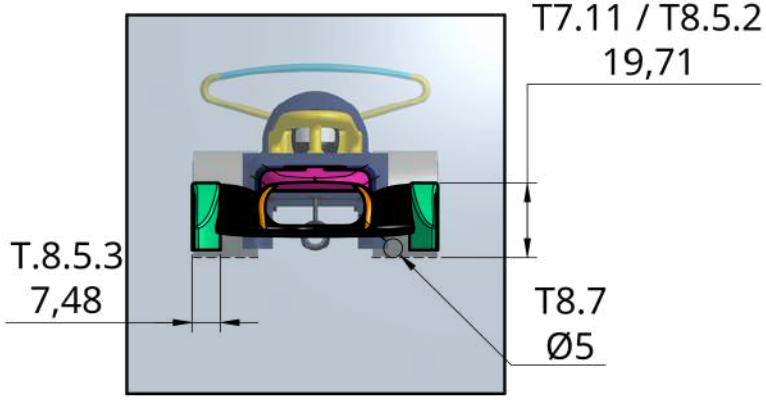
SCALE	WEIGHT	SHEET
1:2		10 of 12



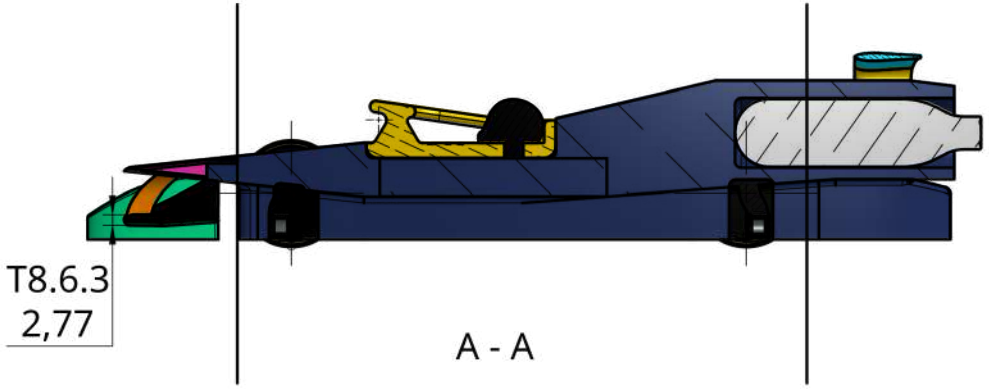
Front Wing - Isometric
1:1



Top



Front



A - A

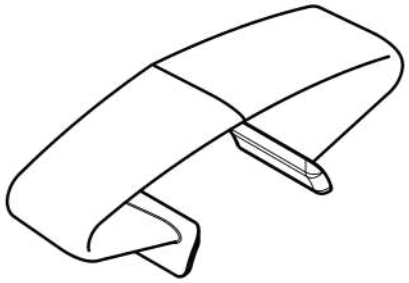
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

Aeris

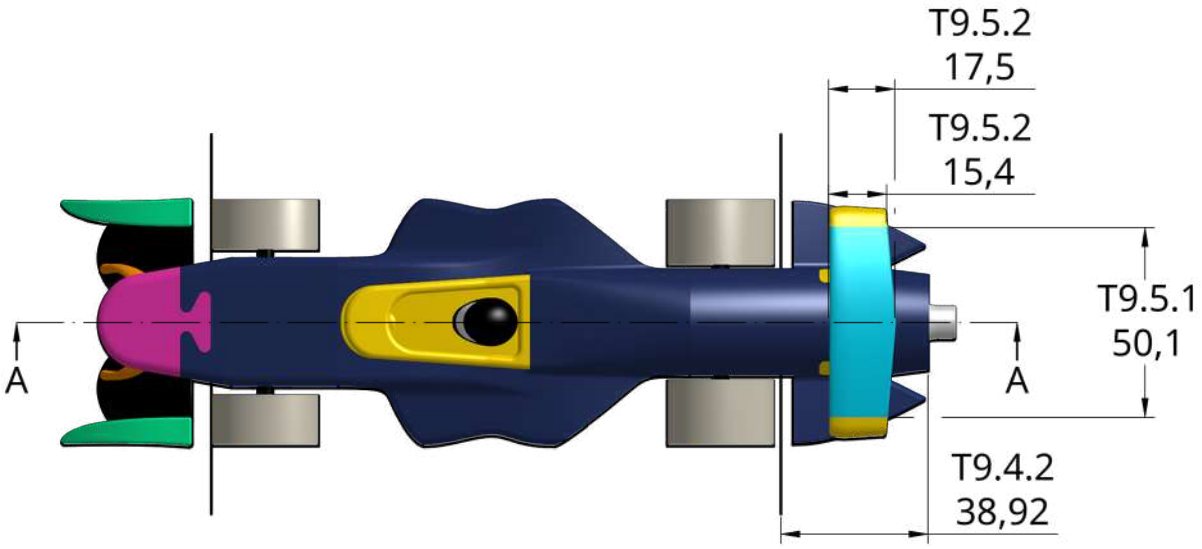
TITLE: Front Wing

SIZE	DWG NO.	REV.
A4		

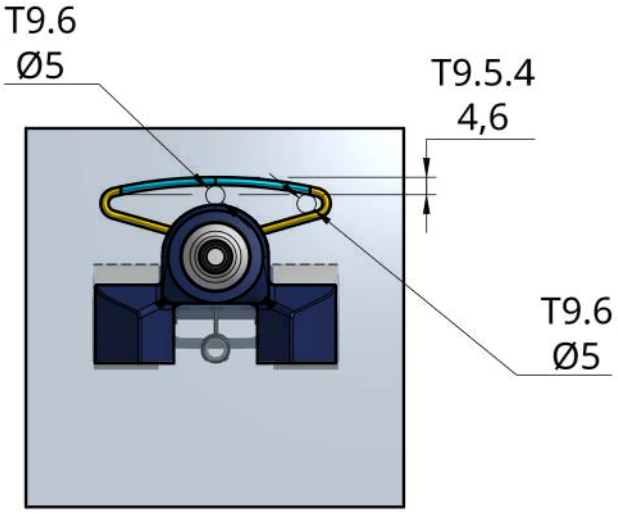
SCALE	WEIGHT	SHEET
1:2		11 of 12



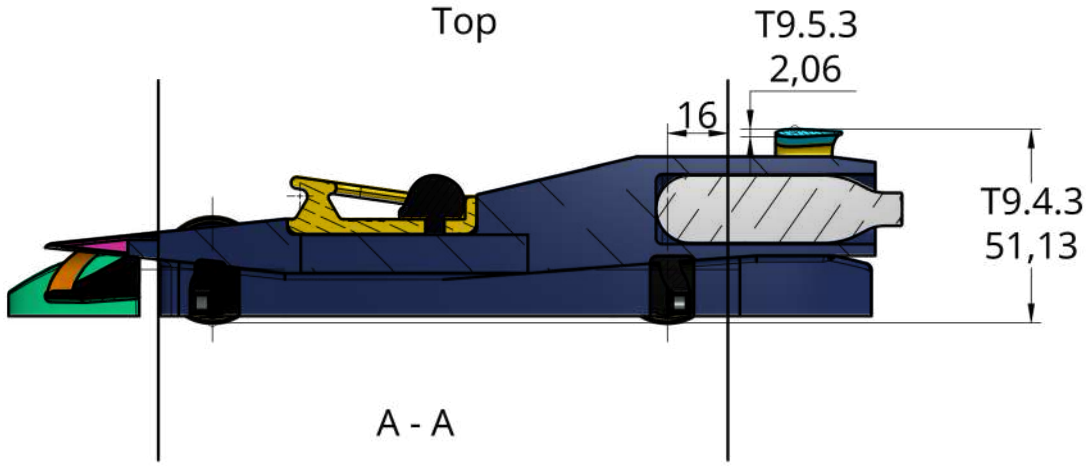
Rear Wing - Isometric
1:1



Top



Back



A - A

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN	KAÉNA TRENCHANT		2025-07-03
SURFACE FINISH	CHECKED			
	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
FIRST ANGLE PROJECTION	MATERIAL		FINISH	

Aeris

TITLE

Rear Wing

SIZE	DWG NO.	REV.
A4		

SCALE	WEIGHT	SHEET
1:2		12 of 12